Fiscal Year:	FY 2017	**Task Last Updated:**	FY 01/17/2017
PI Name:	Volz, Martin Ph.D.		
Project Title:	Reduction of Defects in Germanium-Silicon (RDGS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	martin.volz@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	256-544-5078
Organization Name:	NASA Marshall Space Flight Center		
PI Address 1:	EM31		
PI Address 2:			
PI Web Page:			
City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	03/06/2008	**End Date:**	10/15/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
Contact Email:	jan.r.rogers@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: end date change per P. Downey/MSFC (Ed., 1/12/2015) NOTE: end date change per P. Downey/MSFC (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mazuruk, Konstantin (University of Alabama Huntsville) Cröll, Arne (University of Freiburg)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Original Principal Investigator was Dr. Frank Szofran, Feb 1998-Jan 2004; project continues with Dr. Martin Volz. It is well established that crystals grown without contact with a container have far superior quality to otherwise similar crystals grown in direct contact with a container. In addition to float-zone processing, detached-Bridgman growth is often cited as a promising tool to improve crystal quality, without the limitations of float zoning. In detached growth, most of the melt has contact with the crucible wall, but the crystal is separated from the wall by a small gap, typically 10-100 microns. A liquid meniscus extends between the top edge of the growing crystal and the crucible wall. Detached growth has been found to occur quite often during microgravity experiments and considerable improvements of crystal quality have been reported for those cases. However, no thorough understanding of the process or quantitative assessment of the quality improvements existed prior to the investigation. This project will determine the means to reproducibly grow Ge-Si alloys in the detached mode. Ge-Si alloys are of technological interest because with lower defect density materials it would be possible to make thermal neutron and x-ray optics, faster devices than with silicon, higher efficiency solar cells, and higher thermopower devices. Although these alloys are model materials in the sense that these results should be applicable to many other materials, the major technological significance is that the same production equipment that is currently used for silicon device manufacture can also be used for germanium-silicon devices. This means that using germanium-silicon instead of some other material to achieve improved device performance will save an immense amount of money in development and purchase of production equipment. Several tasks have been completed or are planned as part of the investigation: * measure the relevant material parameters such as contact angle, growth angle, surface tension, and wetting behavior of the GeSi-melt on potential crucible materials; * determine the mechanism of detached growth including the role of convection; * quantitatively determine the differences in defects and impurities among normal Bridgman, detached Bridgman, and float zone (FZ) growth; * investigate the influence of defined azimuthal or meridional flow due to rotating magnetic fields on the characteristics of detached growth; * control time-dependent Marangoni convection in the case of FZ-growth by the use of a rotating magnetic field to examine the influence on the curvature of the solid-liquid interface and the heat and mass transport; and * grow high quality GeSi single crystals with Si concentrations up to 10%. Determine the necessary conditions for reproducibly achieving detachment of Ge-Si alloys in the Bridgman growth method, and quantitatively compare the defect structures and impurity levels of detached Bridgman, normal Bridgman, and floating-zone grown materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our understanding of the mechanisms which lead to crystal detachment and enhance our ability to control detachment both in microgravity and on Earth. Control of detachment will enable the growth of relatively large diameter crystals with orders of magnitude fewer defects than can be achieved with standard growth techniques. The improvement in material quality may lead to better performance for a variety of electronic devices.
Task Progress:	One of the experimental parameters to be varied in the flight experiments is the pressure differential across the meniscus. This meniscus is tens of microns to hundreds of microns in length and bridges the gap between the three-phase-line (solid-liquid-vapor) at the bottom of the crystal and the ampoule wall. Through ground-based testing, ampoule configurations have been developed which will either result in a positive pressure or negative pressure differential across this meniscus. These pressure differences are created by varying the positions of the free gas volumes within the ampoules and the designs will not require changes in the furnace temperature profiles during the crystal growth experiments. Internal ampoule component materials have also been chosen that do not react with other ampoule materials at growth temperatures. The science team has worked with the engineering team to develop a sample cartridge assembly (SCA) configuration that will both survive the anticipated launch environment and enable achievement of the science requirements during flight. An integrated design review (IDR) of the SCA and ampoule configurations is anticipated later in 2017. Our analysis of the dynamic stability of detached growth has been completed and published in the Journal of Crystal Growth 444, 1-8 (2016). The model considered the conditions under which detachment will be stable during crystal growth and is pertinent to the planned silicon-germanium growth experiments planned for the International Space Station (ISS). The stability model analyzed the response of a growing crystal to perturbations to its radius and crystal-melt interface. A linearized differential equation was used to analyze radius perturbations but a linear integro-diffential equation was required for the height perturbations. The stability model was applied to detached solidification under zero-gravity and terrestrial conditions. For terrestrial conditions, a singularity was found to exist in the capillary stability coefficients where, at a critical value of the pressure differential across the meniscus, there is a transition from stability to instability. For the zero-gravity condition, exact formulas for the capillary stability coefficients were derived. Although the stability model was applied to the detached solidification crystal growth method, it can also be advantageously implemented to improve the stability analyses of other shaped crystal growth techniques.
Bibliography Type:	Description: (Last Updated: 07/09/2021)
Abstracts for Journals and Proceedings	Volz MP, Mazuruk K, Croell A. "Detached Solidification of Germanium-Silicon Crystals on the ISS." Presented at the 5th Annual International Space Station (ISS) Research and Development Conference 2016, San Diego, California, July 12-16, 2016. 5th Annual International Space Station Research and Development Conference Abstracts, 2016. https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20160009743.pdf , Jul-2016

Abstracts for Journals and Proceedings	Cröll A, Hess A, Zähringer J, Sorgenfrei T, Senchenkov A, Egorov A, Mazuruk K, Volz MP. "Bridgman Growth of Germanium and Germanium-Silicon Crystals under Microgravity." Presented at the 18th International Conference on Crystal Growth and Epitaxy (ICCGE-18), Nagoya, Japan August 7-12, 2016. 18th International Conference on Crystal Growth and Epitaxy (ICCGE-18), Nagoya, Japan August 7-12, 2016. Abstracts. , Aug-2016
Articles in Peer-reviewed Journals	Mazuruk K, Volz MP. "Dynamic stability of detached solidification." Journal of Crystal Growth. 2016 Jun 15;444:1-8. http://dx.doi.org/10.1016/j.jcrysgro.2016.03.032 , Jun-2016